

Cortical Modulation of Nociception

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Abstract—Nociception is the neuronal process of encoding noxious stimuli and could be modulated at peripheral, spinal, brainstem, and cortical levels. At cortical levels, several areas including the anterior cingulate cortex (ACC), prefrontal cortex (PFC), ventrolateral orbital cortex (VLO), insular cortex (IC), motor cortex (MC), and somatosensory cortices are involved in nociception modulation through two main mechanisms: (i) a descending modulatory effect at spinal level by direct corticospinal projections or mostly by activation of brainstem structures (i.e. periaqueductal grey matter (PAG), locus coeruleus (LC), the nucleus of raphe (RM) and rostroventral medulla (RVM)); and by (ii) cortico-cortical or cortico-subcortical interactions. This review summarizes evidence related to the participation of the aforementioned cortical areas in nociception modulation and different neurotransmitters or neuromodulators that have been studied in each area. Besides, we point out the importance of considering intracortical neuronal populations and receptors expression, as well as, nociception-induced cortical changes, both functional and connectional, to better understand this modulatory effect. Finally, we discuss the possible mechanisms that could potentiate the use of cortical stimulation as a promising procedure in pain alleviation. © 2021 IBRO. Published by Elsevier Ltd. All rights reserved.

Keywords: Nociception, Cerebral cortex, Cortical stimulation, Descending modulation.

INTRODUCTION

Pain is an unpleasant physical and emotional experience associated, or seems to be associated, with a real or potential tissue damage and represents the first cause of clinical consulting around the world (Goldberg and McGee, 2011; Nahin, 2015; Raja et al., 2020). Nociception is the neuronal process of encoding noxious stimuli (Loeser and Treede, 2008). The modulation of nociception-related information can be present at peripheral, spinal and supraspinal levels (Basbaum and Fields, 1978). At supraspinal level, brainstem structures e.g. periaqueductal grey matter (PAG), locus coeruleus (LC), hypothalamic paraventricular nucleus, the nucleus of raphe (RM) and rostroventral medulla (RVM) have been widely studied as part of the endogenous system of nociception modulation through descending modulatory mechanisms at spinal level (Morgan et al., 1989;

Tsuruoka and Willis, 1996; Zhang et al., 1997a, 1997b, 1997c; Zhuo and Gebhart, 1997; Gilbert and Franklin, 2001; Gebhart, 2004; Condés-Lara et al., 2008; Heinricher et al., 2009). At cortical level, Ohara et al., (2005) suggested that modulation of nociception could be performed through two different mechanisms: (i) a descending modulation through activation of brainstem structures; or by (ii) cortico-cortical and cortico-subcortical interactions which could regulate ascending information as well as its processing.

Certainly, clinical studies in humans using neuroimaging techniques showed that when a painful stimuli is applied, several cortical areas are activated, among others, the: insular cortex (IC), anterior cingulate cortex (ACC), prefrontal cortex (PFC), motor cortex (MC) and the primary (SI) and secondary (SII) somatosensory cortices (Treede et al., 1999). Indeed, as discussed below, these areas not only are involved as a final step in the relay of nociceptive information, but also seems to act as an active system enhancing descending modulatory mechanisms that in turn modulates the nociceptive input and consequently the pain experience. In this review, we will discuss evidence about the involvement of these cortices in nociception modulation including the participation of different local neurotransmitters or neuromodulators, besides, the possible suggested hypotheses about the mechanisms of cortical modulation of nociception.

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Abbreviations: ACC, anterior cingulate cortex; IC, insular cortex; LC, locus coeruleus; MC, motor cortex; MCS, motor cortex stimulation; mPFC, medial prefrontal cortex; PAG, periaqueductal grey matter; PFC, prefrontal cortex; RM, nucleus of raphe; RVM, rostroventral medulla; SI, primary somatosensory cortex; SII, secondary somatosensory cortex.

CORTICAL AREAS INVOLVED IN NOCICEPTION MODULATION

Anterior cingulate cortex in nociception modulation

The anterior cingulate cortex (ACC) is connected to many pain-related areas e.g. medial thalamic nuclei, amygdala (AMY), IC and PAG (Neafsey et al., 1993). It is one of the main structures that show increased neuronal activity in presence of nociceptive stimuli (Talbot et al., 1991; Rainville et al., 1997). Indeed, hyperexcitability of this cortex has been related with the pain chronicity under neuropathic condition. For example, Gao et al., (2016) showed that hyperexcitability of the layer V ACC neurons can be recorded in a neuropathic pain model (chronic constriction injury, CCI); in this case this hyperexcitability is primary mediated by an upregulation of the mGluR1 which in turn inhibits the HCN1 channels. Certainly, CCI is associated not only with an increase of neuronal excitability of this cortex, but also with an enhancement of spontaneous neuronal oscillations (Ning et al., 2013; Blomet al., 2014). Similarly, electrical stimulation of ACC as well as its glutamatergic activation induced hyperalgesia in rats to which a thermal noxious stimulus was applied (Calejesan et al., 2000; Zhang et al., 2005). Interestingly, local injection of lidocaine (a blocker of voltage-gated Na⁺ channels) into ACC increased conditioned place preference presumably related to pain relief in animals with chronic pain (Qu et al., 2011).

Together, these data may imply that under a chronic pain condition this cortex is hyperactivated and consequently if we could block the action of excitatory neurotransmitters, a relief of pain could be observed. In this sense, using optogenetics, the activation of specific inhibitory neurons, of which 80% were GABAergic, inside ACC decreased pain scores upon formalin peripheral administration (Gu et al., 2015). Certainly, under the formalin test an increase of glutamate release is observed in the ACC (Xu et al., 2008). Furthermore, it is interesting to note that a possible cortico-thalamic inhibitory mechanism was suggested since the optogenetic stimulation of inhibitory neurons inside ACC decreased neuronal activity in ventroposteromedial (VPM) and ventroposterolateral (VPL) thalamic nuclei upon peripheral pinching (Gu et al., 2015). At this point, we need to keep in mind that an increase of the VPL neuronal activity is associated with a nociceptive input (Yokota, 1989). In this regard, direct ACC projections to the spinal dorsal horn (SDH) at lumbar segments were described, optogenetic inhibition of these projections decreases postsynaptic spinal transmission in a non-RVM dependent mechanism (Chen et al., 2018). Apart from these sensory effects on chronic or inflammatory nociception of this cortex, ACC is believed to mediate also emotional and affective rather than discriminative aspects related to pain perception (Xie et al., 2009; Fuchs et al., 2014; Raison, 2015).

Indeed, although an unspecific electrolytic lesion of this cortex did not have any effect on the spinal nerve ligation (SNL)-induced hypersensitivity, a decrease of the escape-avoidance behavior was observed (LaGraize et al., 2004). These data imply that the ACC play a role in the emotional/affective aspects of chronic pain.

Similarly, Barthas et al. (2015) showed that ACC chemical lesion did not decrease mechanical allodynia-induced by sciatic nerve ligation, however, it decreased the anxiety and depression-like behaviors. Furthermore, Qu et al., (2011) showed that ACC is related to spontaneous aversive response to chronic pain, chemical lesion of ACC (by local injection of ibotenic acid) decreased the place preference related to pain relief, through RVM inhibition, without affecting nerve ligation induced hypersensitivity. Since unspecific lesion of the ACC decrease the emotional/affective aspects of chronic pain and the optogenetic stimulation of channelrhodopsin-2 expressing pyramidal neurons in ACC was associated with anxiety and depression-like behaviors in naïve mice (Barthas et al., 2015), a preponderant role of the ACC on emotional pain aspects is supported.

Regarding the neurotransmitters involved, the participation of different receptors-mediated mechanisms in ACC over nociception modulation has been described (Xie et al., 2009; Fuchs et al., 2014). For example, intracortical or intraperitoneal administration of morphine increased local cortical neuronal currents in ACC induced by peripheral noxious electrical stimulation (Yang et al., 2006). These data contrast previous reports documenting the hyperpolarizing effect induced by activation of μ - and κ -opioid receptors on pyramidal neurons (Tanaka and North, 1994; Giacchino and Henriksen, 1998). In this regard, the antinociceptive effect induced by opioid receptors activation (LaGraize et al., 2006), which are highly expressed in ACC (Mansour et al., 1987), may be associated with modulating local GABAergic activity (Tanaka and North, 1994) or activation of descending modulatory mechanisms (Calejesan et al., 2000). Here, one of the points that could explain these results is the difference in receptor expression among neuronal populations, thus, the increased current may be related to increased intracortical GABAergic neurons activity, as well, pyramidal neurons that mediate a descending mechanism.

Since positron emission tomography (PET) with non-selective opioidergic radioligand, the [¹⁸F]-fluoroethyldiprenorphine showed a high density of opioid receptors in ACC during resting state (Baumgärtner et al., 2006), whereas a decrease of opioid receptors expression was observed in patients with central post-stroke pain (Jones et al., 2004; Willoch et al., 2004), an alteration in endogenous opioid activity in ACC could be related to chronic pain states.

Dopaminergic modulation in ACC over nociception process has been widely studied (Wood, 2008), D₁ and D₂ receptors are expressed inside ACC where D₂ receptors are increasingly expressed in case of neuropathic pain (Ortega-Legaspi et al., 2011). Dopamine microinjection into ACC has been related to decreased autotomy scores (López-Avila et al., 2004). Interestingly, local microinjection of MK-801 (a NMDA receptor antagonist), as well as amantadine (a non-selective dopamine agonist and NMDA receptor antagonist) also produced a significant reduction in autotomy scores. In this regard, the amantadine-induced effect was reversed by prior injection of D₁ and D₂ receptors antagonists (SCH23390 and

spiperone, respectively) suggesting that its antinociceptive effect could be mediated by D₁ and D₂ activation rather than NMDA receptor blockade (López-Avila et al., 2004). Amantadine has also been described to indirectly increase dopamine release as well as dopamine reuptake inhibition (Bailey and Stone, 1975).

Regarding cholinergic transmission in ACC, local scopolamine (a non-selective acetylcholine receptor antagonist) microinjection before a peripheral noxious stimulus decreased autotomy behavior induced by peripheral nerve denervation, however, this effect was absent in case of scopolamine administration posterior to the stimulus administration (Ortega-Legaspi et al., 2003). Additionally, the expression of acetylcholine M₂ receptor in ACC is increased in neuropathic pain conditions (Ortega-Legaspi et al., 2010).

Finally, glycine receptor type A (GlyRA) activation inside ACC through microinjection of taurine, was related to decreased behavioral response against thermal nociceptive stimulus (Pellicer et al., 2007). Briefly, the variation of receptors expression in normal and nociception related conditions, as observed in opioid and dopaminergic receptors, and their specific presence in certain neuronal population within the ACC (μ and κ opioid receptors expression on medial thalamic nucleus afferents) could be of great importance to explain the different effects induced by neurotransmission modulation under different types of noxious stimulation and context-related nociception behavior.

Prefrontal cortex in nociception modulation

Connections from the prefrontal cortex (PFC) to PAG (Hardy and Leichnetz, 1981; An et al., 1998) as well as other pain-related areas e.g. IC, AMY and ventral tegmental area (VTA) have been described (Hadjipavlou et al., 2006; Butler et al., 2011; Mai and Paxinos, 2012). The PFC is activated in presence of noxious stimuli (Lorenz et al., 2003; Apkarian et al., 2005).

Indeed, PFC is believed to participate in nociception modulation at cortical level where it is involved in chronic pain through its connections with IC, and medial thalamic nuclei (Cardoso-Cruz et al., 2013; Henderson et al., 2013; Jurik et al., 2015). Condes-Lara et al., (1991) described that medial prefrontal cortex (mPFC) sustains a tonic facilitatory action over the centralis lateralis nucleus of the thalamus. However, PFC can promote antinociception, as well, by means of its connections with AMY and PAG (Hadjipavlou et al., 2006; Ji et al., 2010; Onozawa et al., 2011; Ong et al., 2019).

As we previously mentioned, the cortical activity could be altered in a different form depending on the type of noxious stimulation. In this context, Dale et al., (2018) described the neuronal activity pattern in PFC in both acute and chronic inflammatory noxious stimulation. While acute pain increased neuronal activity inside PFC, persistent inflammatory pain induced by peripheral administration of complete Freund's adjuvant (CFA) decreased both spontaneous and nociception-evoked neuronal activity (Dale et al., 2018). These data show that persistent pain conditions alter the cortical mechanism of nociception modulation.

Interestingly, a positive linear correlation between basal activity of PFC neurons and pain-evoked firing rates was observed, in other words, decreased basal neuronal activity in PFC (as observed in chronic pain) remodels its capacity to respond upon noxious stimulation, therefore enhancing basal neuronal activity of PFC neurons (through optogenetic stimulation) could improve its modulating mechanism over nociception (Dale et al., 2018). In agreement with this notion, Shiers and Price (2020) reviewed anatomical and circuit changes observed in PFC in chronic pain; which include decreased activity of the projecting pyramidal neurons towards PAG (Cheriyian and Sheets, 2018), morphological changes related to decreased grey matter density (Kuchinad et al., 2007; Geha et al., 2008) and decreased glutamate levels (Kelly et al., 2016; Naylor et al., 2019).

In this regard, repetitive transcranial magnetic stimulation (rTMS) of dorsolateral PFC induced pain relief in patient with fibromyalgia and migraine (Brighina et al., 2004; Conforto et al., 2014; Short et al., 2011). This effect is believed to be mediated by local NMDA and opioid receptors since it was reduced by both ketamine and naloxone (Borckardt et al., 2006; de Andrade et al., 2011, 2014; Taylor et al., 2013; Lamusuo et al., 2017).

Regarding neurotransmitters involved in nociception modulation in PFC, studies in animals and humans reported an increased glutamatergic activity within the mPFC in conditions related to chronic pain (Hung et al., 2014; Kelly et al., 2016; Naylor et al., 2019). For instance, Ji and Neugebauer (2011) described that under normal conditions, the mGlu_{1/5} receptor agonist [(S)-3,5-dihydroxyphenylglycine (DHPG)] decreased spontaneous and stimulus-evoked neuronal activity in mPFC through GABAergic activation since this effect was blocked by bicuculline (a preferent GABA_A receptor antagonist). Similarly, arthritis (a chronic pain condition) decreased both, basal and stimulus-evoked neuronal activity; in this case, the mGlu₁ receptor antagonist [(S)-(+)- α -amino-4-carboxy-2-methylbenzeneacetic acid] administration reversed this effect. Together, these data suggest that the neuronal depression of mPFC present in chronic pain conditions could be mediated by mGlu₁ activation which promotes increased local GABAergic activity. Here, it is worth mentioning that the effect of glutamatergic activation of PFC on nociception modulation could vary on account of the receptor type implicated and the induced net excitatory or inhibitory effect due to the participation of different neuronal populations.

Indeed, ionotropic glutamatergic receptors (NMDA) activation in mPFC produced antinociception (Millecamps et al., 2007; Palazzo et al., 2016), as well, local administration of CX-546 (an AMPA receptor potentiator) increased paw withdrawal latency upon noxious thermal stimulation (Sun et al., 2017). In contrast, increasing activity of metabotropic glutamatergic receptors (e.g. mGlu₅) was associated with pronociception (Kiritoshi and Neugebauer, 2015; Matos et al., 2015; Chung et al., 2017).

These results suggest the presence of different glutamatergic neuronal groups inside mPFC which could explain its dual effect in nociception modulation.

Similarly, GABAergic neurotransmission in PFC promotes nociception or antinociception (Luongo et al., 2013; Zhang et al., 2015) supporting the notion that the observed effect depends on the net activity of different neuronal populations within the PFC.

Accordingly, opioid receptors are highly expressed inside PFC (Jones et al., 1991) and systemic morphine administration decreased the PFC neurons response to local glutamate administration which was reversed by naloxone (Giacchino and Henriksen, 1998). Furthermore, systemic administration of U-69593 (a κ -opioid receptor agonist) decreased synaptic glutamate release inside mPFC induced by basolateral AMY stimulation (Tejeda et al., 2015). On another hand, morphine subcutaneous administration potentiated the antinociceptive effect induced by AMPA/kines in PFC (Sun et al., 2017), suggesting that the result of opioidergic modulation over glutamatergic neurotransmission in PFC is affected by the type of glutamatergic receptor implicated.

In chronic pain conditions, opioid receptors expression is decreased (Willoch et al., 2004; Maarrawi et al., 2007), which could be a result of increased endogenous release or may indicate receptors downregulation or loss of neurons expressing these receptors (Sprenger et al., 2005). In contrast, κ -opioid receptors expression was increased in animals after sciatic nerve constriction (Palmisano et al., 2019). These contrasting observations could be explained by the cortical changes induced by chronic pain, besides, the role of opioid receptors in maintaining long-term potentiation (LTP) implicated in pain chronicity (Bramham and Sarvey, 1996).

Moreover, dopaminergic modulation of nociception in PFC was described (Jarcho et al., 2016; Huang et al., 2019). PFC receives dopaminergic innervations from VTA (Mtui et al., 2012). Also, it is thought to be a site of antinociception induced by VTA stimulation; this effect could be mediated by local D_2 receptors activation (Sogabe et al., 2013).

Noradrenaline (NA) is another neurotransmitter that modulates nociception in PFC which receives direct noradrenergic innervations from LC (Mtui et al., 2012). Contrasting results concerning the effect of NA in PFC over nociception are described: (i) some studies suggest that NA-induced antinociceptive effect is mediated by presynaptic α_1 -adrenoceptors which promotes glutamate release (Zhang et al., 2013; Taylor and Westlund, 2017); whereas (ii) other studies describe that the noradrenergic innervations towards mPFC increase the intensity of spontaneous pain since microinjection of benoxathian (an α_1 -adrenoceptor antagonist) inside mPFC alleviates hypersensitivity induced by trigeminal neuropathy (Kaushal et al., 2016; Hirschberg et al., 2017; Taylor and Westlund, 2017).

Ventrolateral orbital cortex in nociception modulation

Ventrolateral orbital cortex (VLO) receives innervation from the nucleus submedius of thalamus (Sm) and project to PAG which is an important site in nociception modulation (Hardy and Leichnetz, 1981; Craig et al., 1982; Coffield et al., 1992; Yoshida et al., 1992). Electrical stimulation of VLO induced pronociception (Zhang et al.,

1997a, 1997c; Zhang et al., 1997b), meanwhile, other studies showed that GABA administration into this cortex induced antinociceptive effect (Zhang et al., 1998).

Furthermore, VLO is a site of morphine-induced antinociception since local administration of morphine was associated with inhibition of tail flick reflex, and was reversed by naloxone (Huang et al., 2001). Similarly, in visceral nociception, morphine microinjection into VLO attenuated local neurons firing in response to colorectal distention (Yang and Follett, 1998). In this context, it was suggested that VLO promotes a descending modulatory effect at spinal level through PAG activation (Zhang et al., 1997a, 1997c; Zhang et al., 1997b; Huang et al., 2001). Interestingly, PAG lesion or its pharmacological block abolished the VLO-induced inhibitory effect over tail flick reflex (Zhang et al., 1998; Zhang et al., 1997a, 1997c; Zhang et al., 1997b). On the other hand, morphine-induced antinociception in VLO was reversed by local blockade of GABA_A receptors which signals the role of GABAergic activity in this effect (Qu et al., 2006). Again, we should consider different receptors expression could define the net activity and therefore the associated effect.

Dopamine is another neuromodulator that has been studied in VLO on nociception. Local D_1 receptors blockade or D_2 receptors activation were associated with increased pain thresholds, this effect is believed to be mediated by local GABAergic activity (Sheng et al., 2009; Dang et al., 2011).

Moreover, local injection of methoxamine (selective α_1 -adrenoceptors agonist) in VLO attenuates mechanical allodynia induced by spared nerve injury (Wei et al., 2016). Oxytocin produced antinociception when administrated in VLO (Taati and Tamaddonfard, 2018), and finally local activation of 5-HT_{1A} serotonergic receptors by 8-OH-DPAT decreased formalin-induced nociception, an effect mediated by local GABAergic activity that promotes a descending inhibitory modulation, hence it was associated with decreased *c-fos* immunoreactivity at lumbar spinal level (Huo et al., 2010). These results point out the participation of VLO in nociception modulation through descending modulation at spinal level.

Insular cortex in nociception modulation

Insular cortex (IC) is believed to be an important site of physical-emotional integration of pain aspects where its anterior agranular portion has been related to modulate affective and emotional aspects and its posterior granular portion mediates physical aspects of pain (Hanamori et al., 1998a; Mutschler et al., 2012; Denis et al., 2016; Lu et al., 2016). Therefore, we will discuss evidence related to the participation of the anterior and posterior insular cortices in nociception modulation.

Anterior insular cortex in nociception modulation. The rostral agranular insular cortex (RAIC) receives afferents from medial thalamic nuclei (Sm and parvocellular part of the ventral posterior nucleus), which are related to emotional aspects of painful stimuli, besides, it is highly connected with limbic system structures e.g. AMY and

infralimbic cortex (Jasmin et al., 2004). It is generally considered as a pronociceptive cortex since its lesion produced bilateral analgesic effect (Coffeen et al., 2011).

Regarding neurotransmissions modulation in RAIC, changes in local GABA levels inside RAIC could have a dual effect over nociception modulation (pronociception or antinociception) (Jasmin et al., 2003). The authors suggested that RAIC modulates nociception through two different mechanisms (i) RAIC-LC system where glutamatergic innervations from RAIC tonically inhibit LC, thus, local increase in GABA levels disinhibits LC through GABA_A local activity; and (ii) RAIC-AMY system mediated by local GABA_B activity which is believed to inhibit inputs from AMY (a structure known for its pronociceptive effect) (Ansah et al., 2009). Indeed, GABA_B blockade by microinjection of saclofen prior to increasing GABA levels in RAIC induced hyperalgesia, which points out the possible role of RAIC-AMY connection in pronociceptive conditions (Jasmin et al., 2003). In this regard, GABA-induced antinociception in RAIC was reversed by local blockade of GABA_A or intrathecal administration of non-selective noradrenergic receptor antagonist, this shows that GABAergic modulation inside RAIC induces a descending noradrenergic mechanism at spinal level. Here, we can assume that neuromodulators that could induce an increased GABAergic activity inside RAIC could promote antinociception by enhancing a net inhibitory activity inside this pronociceptive cortical area.

Burkey et al., (1996) described RAIC as a site of morphine-induced analgesia where local morphine administration decreased formalin induced nociception through activation of local μ -opioid receptors. This morphine-induced antinociception in RAIC was reversed by systemic administration of naloxone and was associated with decreased *c-fos* reactivity in the spinal dorsal horn at lumbar level after formalin administration suggesting a descending modulatory mechanism (Burkey et al., 1996). However, the direct relation between morphine and GABAergic activity in RAIC on nociception modulation was not explored, an interaction between both elements was observed in the IC in a context related to memory consolidation (Sun et al., 2020).

Furthermore, local inhibition of dopamine reuptake was associated with decreased formalin induced nociception without motor alterations (Burkey et al., 1999). Similar to morphine, the dopaminergic modulatory effect in RAIC is mediated through descending inhibitory mechanisms as it decreased neuronal firing rate of wide dynamic range (WDR) neurons at spinal level upon application of thermal nociceptive stimulus to the skin (Burkey et al., 1999). In this regard, the interaction between dopaminergic receptors activity was described, D₁ receptor blockade or D₂ receptor activation decreased the autotomy score in animals after nerve denervation (Coffeen et al., 2008). Additionally, dopamine levels in the anterior insular cortex was decreased in presence of inflammatory nociception, this was associated with increased D₂ mRNA and decreased D₁ mRNA local expression (Coffeen et al., 2010). Indeed, the involvement of local GABAergic mechanism in dopamine-induced antinociception was described, VTA projections maintain a close relation with

GABAergic neurons in RAIC (Ohara et al., 2003). Finally, the direct effect of GABA receptors inside RAIC on dopamine-induced antinociceptive effect has not been studied, although the differential pattern of dopaminergic receptors expression and the descending modulatory effect signals the importance of considering the neuronal population involved to better understand the induced effect.

Recently, Gamal-Eltrabily et al. (2020) described that oxytocin microinjection in RAIC decreased contralateral hind paw flinches induced by peripheral administration of formalin. This effect is mediated by local oxytocin receptors (OTR) which promote a GABA_A-mediated activity, since oxytocin-induced antinociception was completely reversed by pretreatment with L-368899 (an OTR antagonist) or bicuculline in RAIC. Besides, intrathecal administration of BRL 44408 (an α_{2A} -adrenoceptors antagonist) partially reversed this antinociceptive effect induced by oxytocin in RAIC suggesting the participation of a descending noradrenergic modulatory mechanism. In this study, electrophysiological experiments showed that oxytocin administration in RAIC decreased the biphasic activity of spinal WDR neurons upon peripheral subcutaneous injection of formalin. These results suggest that oxytocin promotes local GABAergic activity in RAIC which triggers a descending inhibitory mechanism at spinal level where the noradrenergic system is partially involved, however, the involvement of other descending mechanisms cannot be excluded (Gamal-Eltrabily et al., 2020).

Additionally, non-steroidal anti-inflammatory drugs (NSAIDs) induced antinociception when injected into the anterior insular cortex. Indeed, decreased NSAIDs induced effect over a 4-day period (NSAIDs tolerance) was mediated by endogenous opioid mechanism (Pirkulashvili et al., 2017). Finally, in an *ex vivo* preparation, nocistatin activated RAIC glutamatergic neurons projecting towards PAG which suggests the involvement of RAIC-PAG-RVM pathway in nociception inhibition (Chen et al., 2010). These data are another evidence of the presence of neuronal population specificity that could explain the dual effect of RAIC over nociception modulation.

Posterior insular cortex in nociception modulation.

Garcia-Larrea (2012) described the posterior insular cortex besides the medial operculum area (PIMO) as a third somatosensory area devoted to processing spinothalamic inputs related to nociception. He mentioned that PIMO has the four necessary characteristics to be considered as a nociception-specific cortical region which include; (i) receives direct innervations from spinothalamic tracts, (ii) is activated in presence of noxious stimulus, (iii) induces nociceptive sensation if directly stimulated, and (iv) if lesioned, it decreases nociception. He claimed that despite not presenting nociception-exclusive processing, this cortex fulfills these characteristics, besides, it has multiple connections with the limbic system, considering that pain is not a special sensation modality, yet, it is an integration system of somatosensory sensation (Moayed and Davis, 2013).

Contrasting results have been documented regarding this proposed hypothesis; meanwhile opercular area was the only cortical area activated by all types of somatosensory stimuli, the posterior insular cortex (PI) showed a specific activation pattern in presence of painful stimuli (Mazzola et al., 2012). Interestingly, patients with isolated posterior insular lesions presented central pain and cold allodynia (increased pain sensation) combined with thermal hypoesthesia (Veldhuijzen et al., 2010). Neurons of this cortex showed an activity pattern (more excitatory than inhibitory) in response to tail pinching in rats (Hanamori et al., 1998a, 1998b), but its electrical stimulation induced a significant decrease in mechanical allodynia after sciatic nerve ligation (Dimov et al., 2018). Moreover, in patient with drug-resistant temporal lobe epilepsy, direct stimulation of posterior insula was associated with noxious as well as non-noxious sensations, suggesting a local somatotopic distribution (Ostrowsky et al., 2002). Taken together, the available data cannot conclude the description of PI as a pain cortex, yet, they show its importance in pain processing.

Therefore, different neurotransmitters have been studied to show the role of this cortex in nociception modulation. In this context, glutamate seems to be the main excitatory neurotransmitter in the IC, as well, it is essential for the long-term potentiation (LTP) induction, a phenomenon believed to be responsible for chronic pain maintenance (Zhuo, 2016).

In PI, local blockade of μ -opioid receptors or cannabinoid-receptor type 1 (CB₁ receptor) abolished the antinociception induced by its electrical stimulation suggesting their participation in this effect (Dimov et al., 2018). Additionally, increased expression of these receptors in other pain-related areas including PAG was described, this signals a possible role of PAG in PI-induced antinociception (Dimov et al., 2018).

Unlike anterior insular cortex, few studies have been performed regarding the role of dopamine in the posterior insula regarding nociception modulation. However, a lower pain activation pattern was observed in PI in patients with Parkinson disease (PD) associated with central primary pain (Brefel-Courbon et al., 2013). Finally, regarding cholinergic neurotransmission in PI, acetylcholine M₂ receptor activation decreased mechanical allodynia in neuropathic pain models (Ferrier et al., 2015).

Motor cortex in nociception modulation

MC has multiple connections with cortical areas involved in nociception process e.g. somatosensory cortex and IC, as well as intralaminar thalamic nuclei and PAG (Wise and Donoghue, 1986; Holstege, 2010). The electrical stimulation of MC was used in the treatment of chronic pain secondary to stroke and refractory to the current pharmacological agents (Tsubokawa et al., 1991; Rasche et al., 2006; Nguyen et al., 2008). Regarding the mechanism involved in the antinociception induced by motor cortex stimulation (MCS), it is not clearly understood how MCS induces analgesia. In any case, we know that, MCS decreased the amplitude of sensory evoked potentials in primary sensory cortex (SI) upon median

nerve stimulation (Kirimoto et al., 2014). Nevertheless, García-Larrea et al. (1999) mentioned that MCS modulates pain through cortico-cortical and cortico-subcortical mechanisms that do not include SI, being the thalamus one of its main regulating points. In support of this notion, MCS was found to inhibit neuronal activity in ventral posterolateral thalamic nucleus (VPL) together with changes in brain activity at the brainstem level suggesting that it modulates both ascending and descending pathways in nociception modulation (Kim et al., 2016). Furthermore, MCS increased cerebral blood flow (CBF), a neuroimaging indicator of increased neuronal activity, in ACC and brainstem, especially in PAG which could be the target of its descending modulatory mechanism (Peyron et al., 2007).

On the other hand, MC could modulate the sensorimotor integration process at spinal level as a part of direct corticospinal projections (Moreno-López et al., 2016; Olivares-Moreno et al., 2017).

These results potentiated the possible use of MCS in pain relief through transcranial stimulation (Fregni et al., 2007). Repetitive transcranial magnetic stimulation (rTMS) of MC was associated with pain relief in patients with neuropathic pain (Lefaucheur et al., 2004a), dystonia (Lefaucheur et al., 2004b), complex regional pain syndrome type I (Pleger et al., 2004) and brachial plexus injuries (Bertolucci et al., 2013). Henssen et al., (2020) reviewed the possible mechanisms involved in antinociception induced by MCS that include: (i) decreased cortical excitability which is NMDA and GABA_A dependent (Hsieh et al., 2012; Muller et al., 2014); (ii) increased dopamine levels in the ventral striatum and nucleus accumbens shell (Loffler et al., 2012; Ohnishi et al., 2004); and (iii) decreased nociception associated with the activation of microglia and astrocytes at spinal level (Kim et al., 2013).

Taken together, the described mechanisms could explain the potent antinociception induced by MCS by modifying cortical neuronal response, as well as, descending modulatory effects which potentiate its use in pain management. However, further studies regarding cortical neuronal populations inside MC and the local participation of different neurotransmitters and their receptors expression on nociception modulation are required.

PRIMARY AND SECONDARY SOMATOSENSORY CORTICES

Though the primary somatosensory cortex (SI) represents a main destiny of sensorial information related to touch, pressure and proprioception also is considered as a part of nociceptive-related structures (Kenshalo et al., 2000; Ploner and Schnitzler, 2007). In this regard, it shows a certain grade of activity in presence of noxious stimulation, besides, it receives afferents from thalamic nuclei e.g. VPL and VPM which are associated with its role in processing certain physical aspects (site, type and intensity) of the noxious stimuli (Willis and Westlund, 1997; Timmermann et al., 2001; Craig, 2003; Ploner and Schnitzler, 2007). Ploner et al., (1999)

reported that a patient who suffered a SI lesion experienced complete loss of tactile and thermal sensation, yet, he described a dull unpleasant sensation (not described as pain) when a specific painful stimulus was applied to the affected hand. Within the SI, neuronal populations in its posterior portion have inputs originating from A δ myelinated fibers activity that encodes what is known as “first or fast pain”, meanwhile, the most anterior portion endures afferent signals originating from unmyelinated C-fibers activity encoding “second or slow pain” (Vierck et al., 2013). This C-fibers related activity in the SI was described as functionally distinct and multimodal, i.e. involved the activity of multiple cortical areas, and maybe important for the spatial recognition of the noxious stimulus which can promote adequate motor response in both humans and rats (Jin et al., 2018).

Here, it is important to consider intracortical neuronal population in SI, to better understand its role in nociception processing. Vierck et al., (2013) provided a detailed review over nociceptive inputs into SI and its intracortical processing. Briefly, nociceptive pyramidal neurons in the area 3a are activated by nociceptive inputs, meanwhile, another group of pyramidal neurons (within the area 3b/1) respond to non-nociceptive afferents. Both groups maintain feedforward excitatory (direct activation) and inhibitory (by GABAergic interneurons) mechanisms, therefore the net effect of the excitatory-inhibitory signals define the neuronal group to be dominantly activated giving rise to the associated sensation (Vierck et al., 2013). This mechanism explains the ability of SI to be activated by both noxious and non-noxious stimuli, besides, its importance in sensory-discriminative aspects of pain processing.

Regarding its role in nociception modulation, case reports have documented the effect of SI lesion on nociception: (i) loss of pain sensation; and (ii) alterations in two-points discrimination with conserved pain sensation or increased pain perception (Marshall, 1951; Bushnell et al., 1999). These “opposing” observations could be attributed to the lesions extension and lack of high-resolution imaging techniques, as suggested by Ploner and Schnitzler (2007). Despite the fact that SI electrical stimulation in rats had no effect on *c-fos* expression in SDH upon peripheral administration of formalin in rats (Gojyo et al., 2002), in humans, transcranial direct current stimulation of SI decreased pain perception (Antal et al., 2008).

Furthermore, local histamine administration into SI produced antinociception which involves the participation of endogenous opioid system (Tamaddonfard and Hamzeh-Gooshchi, 2014). In this context, intrathecal administration of morphine decreased SI neuronal activity upon peripheral thermal noxious stimulation (Kalliomäki et al., 1998).

In chronic pain, an alternation in the intracortical homeostasis or cortical reorganization could be involved in pain persistence (Wrigley et al., 2009), here, restoring the excitatory activity in SI (by selective inhibition of GABAergic neurons) was associated with pain alleviation (Xiong et al., 2017). Nevertheless, increased glutamatergic metabotropic receptor (mGluR3) mRNA expression

was described in the somatosensory cortices in monoarthritic rats (Neto et al., 2001). However, this cortical reorganization was not observed in all types of chronic pain (Gustin et al., 2012). Certainly, SI reorganization has been described after spinal nerve injury (Moore et al., 2000; Ding et al., 2005) and phantom limb phenomenon (Flor et al., 1995).

3Regarding the secondary somatosensory cortex (SII), electrical stimulation was associated with decreased nociceptive behavior and *c-fos* expression in SDH upon peripheral formalin administration (Gojyo et al., 2002). Interestingly, antinociception induced by SII stimulation was reversed by intrathecal administration of Methysergide maleate (serotonin receptor antagonist), but not affected by intraperitoneal administration of naloxone or phentolamine (non-selective α -adrenergic antagonist), which points out the role of descending serotonergic system in this induced effect (Kuroda et al., 2001). In contrast, other studies described the changes in noradrenergic transmission inside SII observed in formalin-induced nociception, besides, increased opioid receptors binding upon induction of spinal LTP by high-frequency stimulation of sciatic nerve (Hjornevik et al., 2010; Nalepa et al., 2010), these results show a possible role of these neuromodulators in SII modulatory effect on nociception.

MECHANISMS OF CORTICAL MODULATION OF NOCICEPTION

Cortical modulation of nociception could be mediated by two mechanisms; (i) descending cortical innervations towards brainstem structures involved in nociception modulation at spinal level, (ii) cortico-cortical and cortico-subcortical interactions (Ohara et al., 2005; Quintero, 2013). In this section, we analyze evidence related to the participation of both mechanisms.

Descending cortical modulation of nociception

As noticed, many cortical areas that modulate nociception-related information project to different brainstem areas or directly to SDH (Hadjipavlou et al., 2006; Moreno-López et al., 2016; Olivares-Moreno et al., 2017; Chen et al., 2018). Indeed, the inhibitory effects induced by stimulation of VLO, mPFC, MC and posterior insular cortex over nociception involves the recruitment of PAG area (Zhang et al., 1997a, 1997b, 1997c; Huang et al., 2001; Peyron et al., 2007; Dimov et al., 2018; Ong et al., 2019).

The PAG does not project directly towards the spinal cord, therefore, it modulates nociception through its projections towards RVM and LC, besides it shows a dual effect on nociception modulation through an “on” and “off” neuronal groups (Gogas et al., 1991; Heinricher and Ingram, 2008; Yamamotová, 2019). Differential glutamatergic activity mediated by mGlu receptors within PAG, as well as, local glutamatergic-GABAergic neuronal populations interactions define the effect of PAG activity over nociception (Maione et al., 2000; Samineni et al., 2017). Therefore, it is noteworthy to take into consideration the specificity of cortical neuronal pop-

ulations that project to PAG and how the experimental procedure affect their activity, together with, the nociception-related cortical changes to better explain the associated cortical effect over nociception (whether pronociception or antinociception).

In this context, opioidergic modulatory systems seems to play a key role in the cortical descending inhibition over nociception (Stamford, 1995; Kharkevich and Churukanov, 1999). In fact, ACC, VLO and RAIC were described as sites of morphine-induced antinociception (Burkey et al., 1996; López-Avila et al., 2004).

Furthermore, LC, the main noradrenergic nucleus, is considered as a site of descending nociceptive modulation (Tsuruoka and Willis, 1996; Zhang et al., 1998a, 1998b; Szabadi, 2012), specially RAIC-induced modulatory effect (Jasmin et al., 2003). RAIC innervations to GABAergic neurons in pericoerulear region tonically inhibit LC in presence of inflammatory nociception, meanwhile, increased GABA levels in RAIC disinhibits LC promoting a descending noradrenergic inhibitory effect at spinal level (Jasmin et al., 2003). In this regard, dopamine and oxytocin were described to promote local GABAergic activity in RAIC, as well, oxytocin-induced antinociception in RAIC is partially promoted by descending noradrenergic mechanism (Ohara et al., 2003; Gamal-Eltrabily et al., 2020). A possible interaction between dopaminergic and oxytocinergic systems in nociception modulation has also been described (Gamal-Eltrabily and Manzano-García, 2018).

Here, we should point out the presence of a feedforward-inhibition performed by RAIC, among other cortical areas, over some brainstem areas (e.g. LC) which maintains nociception response upon the presence of noxious stimulus and could be a possible mechanism of pain chronicity. Considering that chronic pain induces cortical LTP which could potentiate this inhibition and diminishes the endogenous antinociceptive effect, this generates a positive-feedback loop that sustains chronic pain conditions. As noticed, the rupture of this cycle by means of neuromodulation or cortical stimulation could restore the basal cortical activity and disinhibit these brainstem areas.

Finally, the nucleus of raphe (RM) is richly connected with cerebral cortex and could be involved in descending cortical modulation on nociception (Arnsten and Goldman-Rakic, 1984; O'Hearn and Molliver, 1984). Serotonin was also described in the cortical descending inhibitory mechanism (Kharkevich and Churukanov, 1999).

Cortico-cortical and cortico-subcortical modulation of nociception

The advances in neuroimaging allowed a better understanding of pain as an integrated process, not as a somatosensory modality, which is essential to study the neurobiology of this phenomenon (Thompson and Bushnell, 2012; Morton et al., 2016; Inami et al., 2019).

Various techniques e.g. functional magnetic resonance imaging (fMRI), regional cerebral blood flow

(rCBF) and positron emission tomography (PET) have been used to study different brain structures involved in nociception modulation in different physiological, psychological and pathological conditions (Morton et al., 2016).

As mentioned, many cortical and subcortical structures are activated in presence of different modalities of noxious stimuli (Peyron et al., 2000), also, many studies analyzed the interaction between different cortical structures in nociception modulation specially in cognition-related elements. Therefore, when analyzing the role of a certain cortex in nociception modulation, it is essential to be considered as “a part-of-whole” and not as an “isolated” mechanism since cortical areas show high degree of interaction, besides, noxious stimuli induce certain cortical changes (Bornhövd et al., 2002; Rodríguez-Raecke et al., 2009).

In this regard, cortico-cortical interaction could be a means of potentiating the modulatory effect produced by certain cortex. For instance, in rats, RAIC lesion was associated with antinociception in presence of spared nerve injury, this effect was potentiated by MCS (Jung et al., 2016). In humans, electrical stimulation of MC was associated with increased rCBF to insular cortex, ACC, VLO, medial thalamic nuclei and VPL suggesting the participation of these structures in the MCS-induced antinociception (García-Larrea et al., 1999). Another study in animals showed that MCS decreased neuronal activity of VPL and RAIC upon mechanical stimulation suggesting a possible cortico-cortical and cortico-thalamic modulation (Jung et al., 2016). VLO-Sm, RAIC-AMY pathways have also been described as an example of cortico-subcortical mechanisms in nociception modulation (Jasmin et al., 2003; Tang et al., 2009; Yang et al., 2002; You et al., 2013).

Finally, the amplitude of cortical event related potentials, recorded by electrocorticogram in response to peripheral electrical stimulation, were modified by spontaneous or sensory-induced cortical activity meanwhile spinal neurons activity was unchanged, this suggests a local cortical inhibitory mechanism which could be involved in transcranial magnetic or electrical stimulation used for pain alleviation (Béjar-Alonso et al., 2019).

Regarding cortical connectivity in nociception, stimulus-specific functional activity of anterior insular cortex (AI) was correlated with increased activity of ACC and PFC in presence of thermal nociception, meanwhile, PI was functionally associated with somatosensory and primary motor cortices (Peltz et al., 2011). Additionally, AI and ACC were related to nociception modulation during the process of pain anticipation (a phenomenon of decreased or increased pain perception in conditions related to operant learning) (Porro et al., 2002; Lutz et al., 2013). In this context, the blood oxygenation level dependent (BOLD), another index for neuronal activity, can be different within certain cortical areas under different physical and emotional experiences i.e. pain perception (Caria, 2016). Using real-time fMRI, Rance et al. (2014a), Rance et al. (2014b) demonstrated

the importance of dissociation between ACC and PI activity upregulation and downregulation in pain perception.

Furthermore, functional connectivity, measured by computing Pearson's correlation coefficients, between AI and PI and ACC was decreased in patient with complex regional pain syndrome (Kim et al., 2017). Also, decreased cortical activity in areas related to pain inhibition (motor, insular and midcingulate cortices) was observed in patient with fibromyalgia experiencing pain (Glass et al., 2011).

These results suggest the importance of considering the basal state of pain related cortical structures and their connectivity to understand its processing and the associated behaviors. This also could explain the effect of different cognitive conditions e.g. attention and placebo over pain perception since they can modulate the cortical activity and connectivity as well (Moriarty and Finn, 2014; Seminowicz and Davis, 2007).

Regarding cortical changes induced by chronic pain conditions, in patients with chronic lower back pain, a decreased activity of ACC and PFC was observed, which was negatively correlated with pain intensity and perception (Wand et al., 2011). In patients with irritable bowel syndrome, a long-term condition associated with recurrent pain episodes, rectal distention increased the activity of ACC and PFC (Morgan et al., 2005). Indeed, it is believed that LTP-induction at cortical level, as well as cortical remodeling could be possible causes of chronic pain maintenance (Zhuo, 2008; Eto et al., 2011; Moseley and Flor, 2012; Dale et al., 2018). Here, we emphasize the role of possible positive feedback circuits between cortical and subcortical structures (e.g. RAIC-AMY) in sustaining pain chronicity (Jasmin et al., 2003), it will be interesting to analyze the presence of similar circuits between various cortical areas involved in nociception.

Taken together, these results demonstrate the presence of cortical interactions through changing its basal activity in presence of noxious stimuli, as well as induced cortical alterations described in pain conditions that modify nociception processing and modulation.

EXPERIMENTAL LIMITATIONS IN STUDYING CORTICAL

modulation of nociception

One of the main challenges in studying cortical modulation of nociception is the vast variability of the cortical activity in different physiological, psychological, and pathological conditions. Although neuroimaging techniques enabled us to better understand the cortical activation pattern in presence of noxious stimuli and chronic pain conditions, the main limitation is that increased activity described in PET or rCBF does not precisely describe whether it corresponds to excitatory or inhibitory activities. Considering that the cortical neuronal populations show a grade of specificity in their activation pattern as demonstrated in animal models, it is important to distinguish the nature of their increased activity (excitatory or inhibitory). Besides, animal neuroimaging techniques has confirmed some results

obtained in humans about cortical modulation, yet the same activity patterns were not obtained in both species. Besides, controlling animal movements is challenging and the use of anesthesia was associated with results alterations.

Finally, the presence of various cortical and subcortical areas involved in this modulatory process points out the overlapping activity of different modulatory systems and neurotransmitters, this is an experimental challenge since it is difficult to cover all the involved systems in one experimental design taking into consideration the cortical activity patterns observed in different conditions as mentioned.

Pain should be studied as an integrated physical-emotional experience, not as a modality of somatosensory stimulation since it is associated with activation of many structures at cortical and subcortical levels involved in both discriminative and affective aspects. ACC, PFC, VLO, IC, MC and somatosensory cortices represent the main cortical areas involved in modulation of where many neurotransmission systems e.g. glutamate, GABA, opioids, and dopamine (among others) have been studied. The cortical modulation of pain could be mediated by descending activation of brainstem structures e.g. PAG, LC and RVM promoting a descending modulatory effect at spinal effect. However, cortico-cortical, and cortico-subcortical interactions could also be involved in this modulatory effect. Electrical or repetitive transcranial magnetic activation of the cerebral cortex showed good results in pain alleviation which postulates these techniques as promising non-pharmacological procedures in pain management, yet, understanding the mechanisms underlying their effect would be essential for potentiating the results obtained. The overlapping activity of different cortical areas and modulatory systems during nociception is a great challenge to better understand pain processing and modulation at cortical level.

COMPETING INTERESTS

The authors declare no conflict of interest.

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